

Wajeeh Alam

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EDUCATION

University of Waterloo

Sept. 2026 – 2031

Bachelor of Computer Science (Co-op)

Waterloo, ON

- **Relevant Coursework:** Designing Functional Programs (CS 135), Elementary Algorithm Design (CS 136), Algebra (MATH 135), Calculus I (MATH 137)

EXPERIENCE

Senior Computer Instructor

Jun. 2025 - Present

University of Toronto

Toronto, ON

- Engineered interactive digital tools and coding curricula in Python, Scratch & HTML/CSS/JS, delivering programming workshops to 300+ students.
- Deployed interactive teaching games that increased student participation by 40% and improved comprehension of complex logic structures.

Coding & Robotics Instructor

Feb. 2026 – Apr. 2026

MakerKids

Toronto, ON

- Led hands-on coding and robotics lessons for groups of 15+ students, translating programming concepts into Python and Arduino curricula.
- Provided technical equipment & sensors troubleshooting across concurrent sessions, ensuring successful deployment of hardware projects including Snake, Pong and Brick Breaker.

Lead Software Engineer

Jun. 2024 – Sept. 2024

SproutHacks

Waterloo, ON

- Spearheaded the frontend and backend infrastructure for a hackathon registration portal and user dashboard using React, TypeScript, and Tailwind CSS.
- Integrated REST APIs and Firebase authentication to support real-time data syncing and onboarding for 100+ active participants.

PROJECTS

drawOff | *React, TypeScript, Tailwind CSS, Node.js, Socket.io, Git, CI/CD*

May 2026 – Present

- Architected a real-time multiplayer drawing game handling concurrent WebSocket sessions via Socket.io, achieving sub-second state sync across players on a HTML5 Canvas rendering engine.
- Implemented a hybrid judging system where AI vision-model scoring runs by default, with live spectator votes overriding as the winner determinant when present.
- Built matchmaking and room/game-state management handling turn sequencing, round progression, and win conditions across concurrent sessions.

Voca (MathHacks) | *React, TypeScript, Tailwind CSS, Tone.js, MediaPipe*

Feb. 2026

- Engineered a real-time vocal processor and pitch-training platform to train pitch/singing.
- Integrated MediaPipe computer vision for live hand-tracking and autocorrelation algorithms to modulate audio effects and provide gamified pitch feedback.
- Streamlined deployment workflows by implementing CI/CD pipelines and managed version control via Git, placed top 4/60 hackathon projects.

TECHNICAL SKILLS

Languages: Typescript/JavaScript, HTML/CSS, Python, Lua, C++

Frameworks/Libraries: React/React Native, Node.js, Express, SQL/PostgreSQL, Redis, MongoDB, Firebase

Tools/Platforms: Git/GitHub, CI/CD Pipelines